

Renew Power: System Architecture Document for Sustainable Circular Energy Platform

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1 Architecture Overview

Renew Power's system architecture is built on a microservices-based, cloud-native design deployed on AWS with edge processing for real-time 24/7 carbon-free energy (CFE) monitoring, blockchain-based energy passport verification, circular solar panel recycling operations, and corporate energy-transition-as-a-service (ETaaS) management. The architecture is designed to handle the unique challenges of circular energy: real-time hourly matching of renewable generation to corporate demand, blockchain-based REC retirement, IoT-enabled solar panel tracking across their 25+ year lifecycle, carbon credit integration at settlement, and AI-powered grid forecasting across 15 GW of generation assets spanning solar, wind, hydro, and battery energy storage systems (BESS).

The architecture is designed to support 2,000+ corporate customers, 30 GW operational capacity, 500,000+ daily API requests, 5M+ blockchain energy passports issued annually, 500,000 tonnes/year solar panel recycling, and Rs.50,000 crore+ annual revenue by FY30F.

1.1 Design Principles

The architecture follows seven core design principles:

1. **Microservices-first:** Loosely coupled services that can be developed, deployed, and scaled independently. This allows Renew Power to iterate quickly on the 24/7 CFE matching engine without affecting billing or vice versa. Fault isolation ensures that a failure in the notification service does not impact 24/7 CFE matching verification.
2. **Cloud-native:** Leveraging AWS managed services (EKS, RDS, S3, MSK, Lambda, Timestream) for reduced operational overhead. Infrastructure as Code (Terraform) for reproducible deployments. Auto-scaling based on custom metrics.
3. **IoT-first:** Real-time ingestion and processing of telemetry from 50,000+ generation assets (solar inverters, wind turbine controllers, battery management systems, smart meters). Edge processing at generation sites for low-latency anomaly detection.
4. **Blockchain-first:** Every MWh of 24/7 CFE gets an immutable Energy Passport on Hyperledger Fabric, providing tamper-proof proof of hourly matching, REC retirement, and carbon saved. Smart contracts automate REC retirement.
5. **AI-first:** ML models for grid forecasting (generation prediction 0-72 hours), demand forecasting (corporate load curves), curtailment optimization, and solar panel degradation prediction. Models retrained daily.
6. **Security-by-design:** Encryption at rest and in transit, RBAC, comprehensive audit logging for energy transactions (PPA settlements, REC retirements, carbon credits). Network segmentation for SCADA/OT systems.
7. **API-first:** Well-documented RESTful APIs for all external integrations enabling corporate ESG reporting (Salesforce Net Zero Cloud, Persefoni, Watershed) and regulator compliance (CERC, EU CBAM).

1.2 User Personas Supported

The architecture supports eight primary user personas:

Persona	Type	Key Capabilities
Corporate Energy Buyers	B2B	View 24/7 CFE matching dashboard, download hourly REC retirement certificates, access Scope 2 emissions reports, manage ETaaS subscription, initiate solar panel take-back
Grid Operators (DISCOMs)	B2B	Real-time generation data access, scheduled maintenance coordination, deviation settlement mechanism (DSM) reporting
Internal Ops (Generation)	Internal	Monitor 15 GW asset performance, predictive maintenance alerts, curtailment optimization, generation forecasting
Circular Ops (Recycling)	Internal	Solar panel take-back tracking, material recovery (silver/copper/silicon/glass), EPR compliance reporting, blockchain passport updates
Field Service	Internal	Panel inspection scheduling, RFID tag scanning, maintenance work orders, replacement parts ordering
Data API Users	External (corporate)	Aggregated renewable energy data, market insights, ESG reports, carbon credit portfolio
Regulatory Auditors	CERC, EU CBAM, Verra, Gold Standard	Renewable generation verification, REC audit, carbon credit MRV validation, 24/7 CFE compliance reporting
Financial Auditors	Statutory auditors	PPA settlement audit trail, revenue recognition verification (IFRS 15), GST compliance

1.3 High-Level Architecture Layers

Layer	Components
Client Layer	Corporate portal (React.js), mobile app (React Native), internal ops dashboard (Grafana + React), regulatory auditor portal (read-only), API access (REST + GraphQL)
API Gateway Layer	Kong / AWS API Gateway for routing, rate limiting (100-50k req/min), authentication (JWT, API keys, mTLS), observability
Microservices Layer	12+ services including: 24/7 CFE Matching Engine, Energy Passport Engine, ETaaS Billing, Carbon Credit Engine, Recycling Management, Grid Forecasting, Asset Performance Management, Logistics, Analytics, Notification, Fraud Detection
Data Layer	PostgreSQL (operational: customers, PPAs, settlements), TimescaleDB (time-series: generation data), Redis (caching), Snowflake (warehouse), S3 (data lake), Hyperledger Fabric (blockchain: Energy Passport)

External Integrations	Power exchanges (IEX), weather APIs (IMD, ECMWF), payment gateways, carbon registries (Verra, Gold Standard), logistics providers, ESG platforms, grid operator APIs
Deployment & Observability	EKS on AWS (multi-region: ap-south-1, us-east-1, eu-west-2), CI/CD (GitHub Actions + ArgoCD), monitoring (Datadog, Prometheus, ELK, PagerDuty)

2 Client Layer

2.1 Corporate Energy Portal (Web)

Technology: React.js with TypeScript, Tailwind CSS, Next.js for SSR/SEO, WebSocket for real-time data streaming.

Features:

- **24/7 CFE dashboard:** Hourly matching view — renewable generation vs. corporate consumption by location, carbon intensity (gCO/kWh), REC retirement status with blockchain transaction hash.
- **Energy Passport:** Blockchain-verified hourly provenance — generation asset ID, location, timestamp, REC retirement certificate, carbon saved vs. grid baseline.
- **ETaaS subscription management:** View usage vs. contracted PPA, request additional capacity, schedule maintenance windows, download GST-compliant invoices.
- **Carbon offset at settlement:** Real-time residual emissions calculation, one-click offset purchase via Gold Standard credits.
- **ESG reporting:** Automated Scope 2 emissions reports, export to CSV/PDF/JSON-LD, API integration with Salesforce Net Zero Cloud.
- **Solar panel take-back:** Register end-of-life panels for circular recycling, schedule pickup, track material recovery status, view EPR compliance certificates.

2.2 Mobile App (iOS/Android)

Technology: React Native + TypeScript + Redux Toolkit + offline-first SQLite for local persistence.

Features:

- Real-time generation monitoring: View output from specific assets with push notifications for anomalies.
- QR/RFID scan for solar panels: Scan panel QR code → view full lifecycle (manufacturing, installation, degradation, maintenance, recycling status).
- Field service work orders: View assigned tasks, update status, upload photos.
- Offline mode: Cache asset data for last 7 days, sync when connectivity restored.

2.3 Internal Ops Dashboard

Technology: React.js + Grafana (embedded) + WebSocket.

Features:

- Generation asset performance: Real-time SCADA data (MW output, efficiency, availability), predictive maintenance alerts.

- Curtailment monitoring: Real-time curtailment by asset, grid congestion alerts, alternative dispatch recommendations.
- Recycling dashboard: Material recovery rates, EPR compliance status, inventory of recovered materials.

2.4 Corporate API Access

Technology: RESTful APIs (OpenAPI 3.0) + GraphQL + API keys + OAuth2 (client credentials).

Use cases:

- ESG data aggregators: Automated Scope 2 emissions data ingestion (GET /v2/esg/carbon-footprint, /v2/esg/renewable-share).
- Carbon credit buyers: Programmatic offset purchase (POST /v2/carbon/offset), retirement management (POST /v2/carbon/retire).
- Grid operators: Real-time generation data (GET /v2/generation/real-time), scheduled maintenance (PUT /v2/maintenance/schedule).
- Government regulators: Compliance reporting (GET /v2/compliance/cerc), REC audit trail (GET /v2/audit/rec).

3 API Gateway Layer

Component	Specification
Technology	Kong / AWS API Gateway
Authentication	JWT (corporate customers, ops); API keys (B2B, aggregators); OAuth2 (regulators); mTLS (internal services, grid operators)
Rate Limiting	1,000/min (corporate portal); 10,000/min (internal ops); 50,000/min (internal services)
SSL/TLS	TLS 1.3 only; auto-renewal via ACM (90-day)
DDoS Protection	AWS Shield Advanced + WAF (OWASP Top 10, IP reputation)
Request Validation	JSON schema (draft-07); limits: 5MB (corporate), 20MB (internal)
Logging	ELK stack; 90d hot storage, 1y cold (Glacier)
Caching	Redis cluster (5 shards, 3 replicas) for real-time pricing, generation forecasts, REC prices

3.1 API Versioning Strategy

- URI versioning: /v1/generation, /v2/generation.
- Deprecation policy: Announce 6 months in advance; maintain backward compatibility for 12 months.
- Current versions: v1 (stable, 99.95% uptime), v2 (beta: GraphQL support, WebSocket streaming).

4 Microservices Layer

4.1 Microservices Communication Architecture

Services communicate via two primary mechanisms: gRPC for synchronous, high-performance internal calls, and Apache Kafka for asynchronous, durable, event-driven workflows.

gRPC (Synchronous): Used for request-response patterns requiring low latency: 24/7 CFE matching verification during billing, REC retirement validation, inventory lookup for recycling.

Apache Kafka (Asynchronous): Event-driven workflows: when a solar panel is retired, a "panel.retired" event is published → recycling service consumes to schedule pickup, analytics service updates KPIs, blockchain service updates Energy Passport.

4.2 Microservices Summary

Service	Stack	Priority	Function
24/7 CFE Matching Engine	Go + TimescaleDB + Redis	H	Hourly matching of 15 GW generation to 2,000+ corporate load profiles; imbalance settlement
Energy Passport Engine	Go + Hyperledger Fabric + IPFS	H	Blockchain passport issuance (5M+/year), hourly REC retirement, QR code generation, provenance verification
ETaaS Billing Service	Node.js + Stripe + PostgreSQL	H	Subscription billing (Rs./kWh), PPA settlements, revenue recognition (IFRS 15), dunning management
Carbon Credit Engine	Python FastAPI + Gold Standard API	H	Real-time carbon footprint calculation, credit purchase, offset retirement, impact reporting
Recycling Management	Python + PostgreSQL + Redis	H	Solar panel take-back initiation, material recovery tracking, EPR compliance, scrap reconciliation
Grid Forecasting	Python + Prophet + XGBoost	H	Generation forecasting (0-72 hours), load forecasting, price forecasting, curtailment prediction
Asset Performance Management	Go + TimescaleDB + Grafana	H	Real-time SCADA monitoring, predictive maintenance, anomaly detection, solar degradation tracking
Logistics Service	Go + OR-Tools + PostGIS	M	Route optimization (panel pickup), courier allocation, real-time tracking
Analytics & Reporting	Python + dbt + Superset	M	ESG reports (Scope 2), executive KPIs, carbon abatement tracking, REC portfolio management
Notification Service	Node.js + Bull + Resend	L	Multi-channel alerts (email, SMS, push) for generation alerts, maintenance windows
Fraud Detection	Python + XGBoost + Seldon	M	Generation data anomaly detection, REC double-counting prevention, meter tampering detection
Blockchain Verifier	Go + Hyperledger Fabric	H	Energy Passport audit trail, tamper-proof MRV, smart contracts for REC retirement

4.3 Service 1: 24/7 CFE Matching Engine

Handles hourly matching of renewable generation to corporate demand.

Technology: Go 1.21 + TimescaleDB + Redis + Kafka.

Key functions:

- **Hourly matching:** For each corporate buyer, match actual renewable generation against actual consumption. Calculate surplus/deficit. Target: 99% matching rate.
- **Imbalance settlement:** If generation < consumption, purchase RECs from IEX. If generation > consumption, bank excess RECs or sell to other buyers.
- **Real-time pricing:** Retrieve IEX DAM/GDAM prices every 15 minutes via API. Cache in Redis.
- **Blockchain integration:** For every matched hour, publish event to Kafka → Energy Passport Engine creates blockchain record.

4.4 Service 2: Energy Passport Engine

Handles blockchain-based energy verification and REC retirement.

Technology: Go 1.21 + Hyperledger Fabric 2.5 + IPFS + PostgreSQL.

Key functions:

- **Passport issuance:** Every MWh of 24/7 CFE gets immutable Energy Passport. Data stored on blockchain: passport ID, asset ID, location, timestamp, REC hash, carbon saved.
- **QR code generation:** Each passport has unique QR code for verification.
- **Verification:** Customer scans QR code → API calls blockchain node → returns passport data in <500ms.
- **Circular journey updates:** When solar panel is retired and recycled, new transaction added: panel ID, retirement date, material recovery quantities.
- **Smart contracts for REC retirement:** Validate REC not double-retired, mark as retired, create Energy Passport, emit event.

4.5 Service 3: ETaaS Billing Service

Handles corporate subscription billing and PPA settlements.

Technology: Node.js 20.x + Stripe Connect + PostgreSQL + Redis.

Key functions:

- **Subscription lifecycle:** Signup (ReNew Verified Rs.3.5/kWh, ReNew Carbon-Neutral Rs.3.75/kWh, ReNew Circular Rs.4.0/kWh), pause, resume, cancel.
- **Usage-based billing:** Read consumption from smart meter (hourly). Apply contracted rate. Generate monthly invoice (GST-compliant).
- **PPA settlements:** For fixed-quantity PPAs, calculate monthly settled volume vs. contracted. Apply penalties per CERC DSM rules.

- **Revenue recognition:** IFRS 15 compliant. Revenue recognized as energy is delivered.

4.6 Service 4: Carbon Credit Engine

Handles real-time carbon footprint calculation and offset purchases.

Technology: Python FastAPI + Gold Standard API + Verra API + Redis.

Key functions:

- **Real-time carbon footprint:** Calculate residual emissions = (MWh consumed × grid CI) - (MWh from Renew × Renew CI).
- **Credit sourcing:** Bulk purchase of Gold Standard credits (5 million tonnes/year at \$8–12/tonne).
- **Offset retirement:** Call Gold Standard/Verra API to retire credits on customer's behalf. Generate retirement certificate (PDF + JSON-LD).
- **Corporate offtake:** Manage 5-10 year offtake agreements with corporate buyers. Track delivery against annual quotas.

4.7 Service 5: Recycling Management Service

Handles solar panel take-back and material recovery tracking.

Technology: Python + PostgreSQL + Redis + Celery + Shiprocket API.

Key functions:

- **Take-back initiation:** Customer registers end-of-life panels (QR scan or serial number). AI fraud detection (image comparison, serial number validation).
- **EPR credit calculation:** Calculate compliance credit based on panel age, recovery rates achieved.
- **Material recovery tracking:** Panel received → RFID scan → automated sorting → material recovery (silver 95%, copper 92%, silicon 85%, glass 90%). Blockchain update with recovery quantities.
- **Scrap reconciliation:** Recovered materials sold on commodity markets (LBMA for silver, LME for copper). Revenue reconciled against EPR liability.

4.8 Service 6: Grid Forecasting Engine

Predicts renewable generation, load, and prices for 0-72 hours.

Technology: Python + Prophet + XGBoost + TimescaleDB + Grafana.

Key functions:

- **Generation forecasting (solar):** Ensemble of LSTM (0-6 hours), transformer (6-48 hours), physical models. Accuracy: 92–95%.
- **Generation forecasting (wind):** Features: NWP, historical generation, turbine degradation curves. Accuracy: 88–92%.

- **Load forecasting:** Features: historical consumption, day-of-week, holidays, temperature. Accuracy: 85%.
- **Price forecasting:** IEX DAM/GDAM price prediction. Features: generation forecast, load forecast, fuel prices. Accuracy: 80%.
- **Curtailement prediction:** Identify grid congestion events 24-48 hours ahead. Recommend alternative dispatch.

4.9 Service 7: Asset Performance Management

Monitors 15 GW generation assets in real time.

Technology: Go + TimescaleDB + Grafana + Kafka.

Key functions:

- **Real-time SCADA:** Ingest telemetry from 50,000+ assets every 5 seconds. 5M+ readings/second at peak.
- **Predictive maintenance:** ML models trained on historical failure data. Predict failures 3 months in advance with 85% accuracy.
- **Solar panel degradation:** IV curve monitoring from 500,000+ panels. Predict remaining useful life within ± 6 months.
- **Anomaly detection:** Isolation Forest on generation data. Detect underperformance, meter tampering, communication failures.

4.10 Service 8: Logistics Service

Manages forward logistics and reverse logistics.

Technology: Go + Google OR-Tools + PostGIS + Redis.

Key functions:

- **Courier allocation:** Select optimal courier partner based on cost, delivery time, success rate, pin code coverage.
- **Route optimization:** For panel pickup from corporate sites, solve Vehicle Routing Problem (VRP). Minimize total distance, respect time windows.
- **Real-time tracking:** Integrate with courier APIs. Customer-facing dashboard with live location, ETA predictions.
- **Smart scheduling:** Assign work orders to field service technicians based on proximity, skillset, availability.

5 Data Layer

5.1 Operational Databases

Database	Technology	Purpose	Size (FY30F)
Primary OLTP	PostgreSQL 15	Customers (corporate), PPAs, settlements, invoices, contracts	20 TB
Time-series	TimescaleDB	Generation telemetry (5M readings/second), load consumption, pricing data	100 TB
Blockchain	Hyperledger Fabric 2.5	Energy Passport audit trail, immutable records, cryptographically linked	50 TB
Caching	Redis Enterprise	Real-time pricing, generation forecasts, REC prices, session management	500 GB
Geospatial	PostGIS	Asset locations, grid topology, pickup/delivery routing	5 TB
Search	Elasticsearch 8.x	Customer search, asset search, audit log search	3 TB

5.2 Data Warehouse

Technology: Snowflake with auto-scaling (X-Small to 4X-Large).

Data sources: All microservices stream data to Kafka (50+ topics) → transformed via dbt (150+ models) → loaded to Snowflake.

Data volume: 10 TB raw data per month (FY26E) → 100 TB by FY30F.

Refresh frequency: Real-time for operational dashboards; daily for analytical reports.

5.3 Data Lake

Technology: Amazon S3 (intelligent-tiering) + Glacier Deep Archive.

Content:

- Satellite imagery (cloud cover, solar irradiance): Retained 30 days (hot), 1 year (cold)

- SCADA telemetry (historical): Retained 1 year (hot), 10 years (cold) — CERC regulatory requirement
- Customer contracts (PPAs, offtake agreements): Retained 15 years post-expiry
- Blockchain audit trail: Retained indefinitely (Verra/Gold Standard requirement)
- System logs: Retained 30 days S3 Standard, 90 days Glacier Instant, 1 year Glacier Deep Archive
- ML model artifacts: Versioned, retained indefinitely for reproducibility

6 Scalability & Performance

6.1 Scaling Strategy

The architecture is designed to scale horizontally to handle 5x growth by FY30F.

Component	FY26E	FY30F
Corporate customers	500	2,000
Operational capacity (GW)	15	30
Daily API requests	1,000,000	10,000,000
Hourly matching records (daily)	36,000	144,000
Solar panels tracked for recycling	500,000	5,000,000
Data warehouse size (TB)	10	100
Concurrent WebSocket connections	1,000	10,000

6.2 Performance SLAs

Service	Target (p95)	Current (p95)
24/7 CFE matching verification	1 sec	600 ms
Energy Passport verification (QR scan)	500 ms	300 ms
Generation forecast (72-hour)	5 sec	3 sec
Invoice generation (monthly)	10 sec	5 sec
Solar panel take-back initiation	2 sec	1 sec
Material recovery tracking update	1 sec	500 ms
Real-time generation dashboard	200 ms	100 ms

7 Security Architecture

7.1 Authentication & Authorization

User Type	Authentication Method & Controls
Corporate Customers (Portal)	JWT + password (bcrypt hashed, 12 rounds). MFA optional. 8 hr timeout. 5 failed attempts → 15 min lock-out.
Corporate API (B2B)	API keys + OAuth2 (client credentials) + IP whitelist. Key rotation 90 days. Scope-limited permissions.
Internal Ops (Employees)	JWT + MFA (TOTP) mandatory. 12 hr timeout. Device attestation.
Field Service (Mobile)	JWT + biometric (fingerprint/Face ID). Offline token (cached, 7 day expiry).
SCADA/OT (Industrial)	mTLS + client certificates + network segmentation (isolated VPC). No internet access.
Grid Operators (DISCOMs)	OAuth2 (authorization code) + client certs. Read-only access. Session timeout 2 hours.
Regulatory Auditors	OAuth2 + client certs. Read-only access to audit trail.
Internal Services	mTLS (Istio). SPIFFE identities. Auto cert rotation (24 hours).

7.2 RBAC Roles and Permissions

Role	Permissions
Corporate Customer	View own 24/7 CFE matching, Energy Passport, invoices, ESG reports; download REC certificates; initiate panel take-back.
Internal Ops (Generation)	View all generation assets, forecasts; create work orders; update asset metadata. No access to financial data.
Internal Ops (Recycling)	View take-back queue; update recovery tracking; manage EPR compliance; update blockchain.
Field Service	View assigned work orders; update status; upload photos. No access to other regions' data.
Finance	View all invoices, PPA settlements; manage billing; run revenue reports. No access to SCADA.
Admin	User management; system configuration; report access; audit log viewer.
Super Admin	Full system access (break-glass only, requires 2-person approval); audit log review; DR failover authorization.

8 Conclusion

The Renew Power system architecture is designed to support the company's ambitious growth targets from Rs.22,000 crore (FY26E) to Rs.50,000+ crore (FY30F) while maintaining high availability (99.95% uptime target), sub-second response times for critical operations (24/7 CFE matching verification, Energy Passport scans), and full compliance with CERC grid codes, EU CBAM, India's Carbon Credit Trading Scheme (CCTS), and EPR regulations for solar panel recycling.

8.1 Key Architectural Decisions Summary

Decision	Rationale
Microservices (vs. monolithic)	Independent scaling (24/7 CFE matching vs. billing vs. recycling). Team autonomy. Fault isolation.
EKS (K8s) over ECS	Portability (multi-cloud for EU data sovereignty), HPA (custom metrics), ecosystem (CNCF, Helm).
PostgreSQL + TimescaleDB	Single DB for transactional + time-series. TimescaleDB compression (90% space savings).
Hyperledger Fabric for blockchain	Permissioned (approved validators only), high throughput (1,000+ TPS), privacy (channels).
Multi-region deployment	Low-latency for India, US, Europe. Data residency (GDPR, India DPDP). DR protection.
React Native for mobile	Single codebase (iOS/Android), consistent features, offline support.

8.2 Projected Capacity by FY30F

- Corporate customers: 2,000 (4x growth from 500)
- Operational capacity: 30 GW (2x growth from 15 GW)
- Daily API requests: 10,000,000 (10x growth from 1M)
- Hourly matching records (daily): 144,000 (4x growth from 36,000)
- Energy Passport scans (monthly): 5,000,000 (10x growth from 500K)
- Solar panels tracked for recycling: 5,000,000 (10x growth from 500K)
- Material recovery volume (tonnes/year): 500,000 (10x growth)
- Blockchain transactions (daily): 200,000 (10x growth from 20K)
- Data warehouse size: 100 TB (10x growth from 10 TB)
- Annual revenue: Rs.50,000 crore (2.3x growth from Rs.22,000 crore)

8.3 Architecture Readiness Assessment

The architecture is projected to handle these targets with:

- **Uptime:** 99.95% (4.38 hours downtime/year), with 99.99% target for 24/7 CFE matching and Energy Passport verification (52 minutes/year)
- **P95 latency:** 1 sec for critical APIs (matching verification, passport scan, invoice generation)
- **RTO (Recovery Time Objective):** 2 hours (DR), with 1 hour RTO for CFE matching and billing services
- **RPO (Recovery Point Objective):** 15 minutes (5 minutes for critical data: PPAs, settlements, blockchain passports)
- **Scalability:** Horizontal scaling verified by load tests (10x current load)